Lecture 5. Continuous Random Variables

Densities, uniform, normal, exponential, hazard rates, gamma and beta, transformations

Contents

5.1 Densities **2**

5.2 Expectation and variance **2**

5.3 The uniform distribution **3**

5.4 Normal random variables **3**

5.5 Exponential random variables **4**

5.6 Other continuous distributions **4**

5.7 The distribution of a function of a random variable **5**

5.8 Selected exercises **5**

5.1 Densities

X is **continuous** if $P(X \in B) = \int_B f(x)\,dx$ for a nonnegative **density** f and all (reasonable) B. Then $\int f = 1$, $P(X = a) = 0$, $F(a) = \int_{-\infty}^{a} f$, $F' = f$ where f is continuous, and $P(a - \frac{\epsilon}{2} \le X \le a + \frac{\epsilon}{2}) \approx \epsilon f(a)$.

Example 5.1

(a) $f(x) = C(4x - 2x^2)$ on $(0, 2)$: $1 = 8C/3$, so $C = 3/8$, and $P(X > 1) = 1/2$ by symmetry.
(b) If $Y = 2X$ then $F_Y(a) = F_X(a/2)$ and $f_Y(a) = \frac{1}{2} f_X(a/2)$.

5.2 Expectation and variance

Motivation. Sums become integrals; the tail formula $E[X] = \int_0^\infty P(X > x)dx$ is often quickest.

$E[X] = \int x f(x)\,dx$; $\mathrm{Var}(X) = E[X^2] - \mu^2$.

Lemma 5.1 (Tail formula)

If $Y \ge 0$, then $E[Y] = \int_0^\infty P(Y > y)\,dy$.

Proof. $\int_0^\infty P(Y > y)\,dy = \int_0^\infty \int_y^\infty f_Y(x)\,dx\,dy = \int_0^\infty \Big(\int_0^x dy\Big) f_Y(x)\,dx = \int_0^\infty x f_Y(x)\,dx.$ ■

(Discrete analogue: $E[N] = \sum_{k \ge 1} P(N \ge k)$, $N \ge 0$ integer.)

Proposition 5.2

$E[g(X)] = \int g(x) f(x)\,dx$.

Proof. For $g \ge 0$, by the lemma, $E[g(X)] = \int_0^\infty P(g(X) > y)\,dy = \int_0^\infty \int_{x:g(x)>y} f(x)\,dx\,dy = \int \Big(\int_0^{g(x)} dy\Big) f(x)\,dx = \int g(x) f(x)\,dx$. In general split $g = g^+ - g^-$. ■

Example 5.2 (Breaking a stick)

A unit stick is broken at a uniform U. The piece containing a fixed $p \in [0, 1]$ has length $L_p(U) = 1 - U$ $(U < p)$ or U $(U > p)$:

$$E[L_p(U)] = \int_0^p (1 - u)\,du + \int_p^1 u\,du = \frac{1}{2} + p(1 - p).$$

This is $\ge \frac{1}{2}$, maximal at $p = \frac{1}{2}$: longer pieces are more likely to contain p (cf. Example 4.3).

Example 5.3 (When to leave)

Travel time X has density f, cdf F; arriving s early costs cs, s late ks. Leaving t minutes ahead costs on average

$$E[C_t(X)] = \int_0^t c(t - x) f(x)\,dx + \int_t^\infty k(x - t) f(x)\,dx, \qquad \frac{d}{dt} E[C_t(X)] = cF(t) - k\{1 - F(t)\}.$$

The optimum is $F(t^*) = k/(k + c)$, the median for $k = c$; likewise $E|X - a|$ is minimized at

the median.

5.3 The uniform distribution

$X \sim U(\alpha, \beta)$ has density $1/(\beta - \alpha)$ on (α, β). Then $E[X] = (\alpha + \beta)/2$ and $\mathrm{Var}(X) = (\beta - \alpha)^2/12$.

Example 5.4 (Bertrand's paradox)

Is a "random chord" of a circle of radius r longer than the side $\sqrt{3}r$ of the inscribed triangle? A chord at distance D from the centre has length $2\sqrt{r^2 - D^2} > \sqrt{3}r$ iff $D < r/2$.
(i) $D \sim U(0, r)$ (perpendicular through a random point of a radius): $1/2$.
(ii) One endpoint fixed, angle to the tangent $\theta \sim U(0, \pi)$: length $2r \sin\theta > \sqrt{3}r$ iff $\theta \in (\pi/3, 2\pi/3)$: $1/3$. "Random chord" is undefined until the experiment is specified.

5.4 Normal random variables

$X \sim N(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \qquad -\infty < x < \infty.$$

It integrates to 1 since $I^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\,dr\,d\theta = 2\pi$, $I = \int e^{-x^2/2}\,dx$. Since $aX + b \sim N(a\mu + b, a^2\sigma^2)$, $Z = (X - \mu)/\sigma$ is **standard normal** (cdf Φ, $\Phi(-x) = 1 - \Phi(x)$), with $E[Z] = 0$ and $E[Z^2] = 1$ (by parts); so $E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, and

$$P(X \le a) = \Phi\Big(\frac{a - \mu}{\sigma}\Big).$$

$P(|Z| < 1, 2, 3) \approx 0.683, 0.954, 0.997$; $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$; $1 - \Phi(x) \approx e^{-x^2/2}/(x\sqrt{2\pi})$ for large x.

Example 5.5 (Detecting a signal)

A bit is sent as ± 2, received as $R = \pm 2 + N$, $N \sim N(0, 1)$, decoded as 1 iff $R \ge 0.5$. Error probabilities: $P(N < -1.5) \approx 0.067$ and $P(N \ge 2.5) \approx 0.006$.

Theorem 5.3 (De Moivre–Laplace)

If $S_n \sim \mathrm{Bin}(n, p)$, then for $a < b$, $P\Big(a \le \dfrac{S_n - np}{\sqrt{np(1-p)}} \le b\Big) \to \Phi(b) - \Phi(a)$ as $n \to \infty$.

A special case of the CLT (Lecture 8), good when $np(1-p) \ge 10$. Use the **continuity correction** $P(S_n = i) = P(i - \frac{1}{2} < S_n < i + \frac{1}{2})$; for small p use Poisson.

Example 5.6

(a) 40 fair flips: $P(S = 20) \approx \Phi(0.5/\sqrt{10}) - \Phi(-0.5/\sqrt{10}) \approx 0.1256$ (exact 0.1254).
(b) 450 admitted, each enrolling with probability 0.3, room for 150: $S \sim \mathrm{Bin}(450, 0.3)$, mean 135, variance 94.5, $P(S \ge 151) = P(S \ge 150.5) \approx 1 - \Phi(15.5/\sqrt{94.5}) = 1 - \Phi(1.59) \approx 0.055$ (exact: 0.057).

5.5 Exponential random variables

$X \sim \mathrm{Exp}(\lambda)$: $f(x) = \lambda e^{-\lambda x}$, $P(X > x) = e^{-\lambda x}$, $x \geq 0$; $E[X] = 1/\lambda$, $\mathrm{Var}(X) = 1/\lambda^2$. In a Poisson process (Example 4.7) the first event comes after t iff $N(t) = 0$, probability $e^{-\lambda t}$.

Proposition 5.4 (Memorylessness)

$P(X > s + t \mid X > t) = P(X > s)$ for all $s, t \geq 0$ holds for $X \sim \mathrm{Exp}(\lambda)$, and the exponential distributions are the only continuous distributions on $[0, \infty)$ with this property.

Proof. $e^{-\lambda(s+t)}/e^{-\lambda t} = e^{-\lambda s}$. Conversely $\bar{F}(x) = P(X > x)$ satisfies $\bar{F}(s + t) = \bar{F}(s)\bar{F}(t)$, so $\bar{F}(m/n) = \bar{F}(1/n)^m$, $\bar{F}(1) = \bar{F}(1/n)^n$, hence $\bar{F}(x) = \bar{F}(1)^x$ for rational x and, by right continuity, all $x \geq 0$: $\bar{F}(x) = e^{-\lambda x}$, $\lambda = -\log \bar{F}(1)$. ■

Example 5.7 (Two clerks)

A customer arrives while Jones and Brown are being served by two clerks and is served when either finishes; service times are i.i.d. exponential. When the first leaves, the other's remaining time is again exponential (memorylessness), so the newcomer is last with probability 1/2.

Hazard rate. For a lifetime X, $\lambda(t) = f(t)/\bar{F}(t)$ ($\lambda(t)\,dt$: failure probability in dt at age t). Since $\lambda(t) = -\frac{d}{dt}\log \bar{F}(t)$,

$$\bar{F}(t) = \exp\Big\{-\int_0^t \lambda(s)\,ds\Big\},$$

so λ determines F; constant hazard is exponential, increasing hazard models ageing.

Example 5.8

If a smoker's death rate is twice a non-smoker's at every age, the smoker's survival from A to B is $\exp\{-2\int_A^B \lambda_N\}$, the *square* of the non-smoker's: 0.76 (age 50 to 60) becomes 0.578.

5.6 Other continuous distributions

Gamma. $f(x) = \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}/\Gamma(\alpha)$, $x > 0$, with $\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\,dy = (\alpha - 1)\Gamma(\alpha - 1)$, $\Gamma(n) = (n - 1)!$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$; $E[X] = \alpha/\lambda$, $\mathrm{Var}(X) = \alpha/\lambda^2$. The n-th event time T_n of a Poisson process is Gamma(n, λ):

$$P(T_n \leq t) = P(N(t) \geq n) = \sum_{j \geq n} e^{-\lambda t}\frac{(\lambda t)^j}{j!},$$

whose derivative in t telescopes to the density $\lambda e^{-\lambda t}(\lambda t)^{n-1}/(n - 1)!$. **Chi-squared** with n d.f. is Gamma$(n/2, 1/2)$.

Weibull. $F(x) = 1 - \exp\{-((x - \nu)/\alpha)^\beta\}$, $x > \nu$; weakest-link lifetimes, hazard $\propto (x - \nu)^{\beta-1}$.

Cauchy. $f(x) = \frac{1}{\pi}\frac{1}{1+(x-\theta)^2}$. A flashlight at distance 1 from a wall at uniform angle $\theta \in (-\pi/2, \pi/2)$ lights $X = \tan\theta$, with $P(X \leq x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$. No mean: $\int |x|/(1 + x^2)\,dx = \infty$.

Beta. $f(x) = x^{a-1}(1-x)^{b-1}/B(a,b)$ on $(0,1)$, $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $E[X] = a/(a+b)$, $\mathrm{Var}(X) = ab/\{(a+b)^2(a+b+1)\}$; random proportions ($a = b = 1$: uniform).

Pareto. $Y = ae^X$, $X \sim \mathrm{Exp}(\lambda)$, has $P(Y > y) = (a/y)^\lambda$, $y \geq a$; given $Y > y_0$ it is Pareto with $a \to y_0$.

5.7 The distribution of a function of a random variable

Motivation. The density of $Y = g(X)$: differentiate the cdf; for monotone g this gives a Jacobian formula.

Theorem 5.5

If g is strictly monotone and differentiable on the range of X, then $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

for y in the range of g, and 0 otherwise.

Proof. Increasing g: $F_Y(y) = F_X(g^{-1}(y))$; differentiate. Decreasing: $F_Y(y) = 1 - F_X(g^{-1}(y))$ and $(g^{-1})' < 0$. ■

Example 5.9

(a) $Y = X^2$: $F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$, $f_Y(y) = \frac{1}{2\sqrt{y}}\{f_X(\sqrt{y}) + f_X(-\sqrt{y})\}$; for $X = Z$, $\frac{1}{\sqrt{2\pi y}} e^{-y/2}$, the χ^2_1 density (so $\Gamma(\frac{1}{2}) = \sqrt{\pi}$).
(b) $|X|$ has density $f_X(y) + f_X(-y)$ for $y > 0$.
(c) Lognormal: $Y = e^X$, $X \sim N(\mu, \sigma^2)$, $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma y} \exp\{-(\log y - \mu)^2/(2\sigma^2)\}$, $y > 0$ (multiplicative price changes).
(d) For continuous strictly increasing F, $F(X) \sim U(0,1)$ and $F^{-1}(U) \sim F$: the basis of simulation.

5.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): tail moments, a variance bound, normal moments, and the probability integral transform for general F.

Exercise 5.5 (Moments from the tail)

Use Lemma 5.1 to show that for a nonnegative random variable X and $n \geq 1$,

$$E[X^n] = \int_0^\infty n x^{n-1} P(X > x)\, dx.$$

Deduce $E[X^2]$ for $X \sim \mathrm{Exp}(\lambda)$.

Solution.

Step 1. By Lemma 5.1, $E[X^n] = \int_0^\infty P(X^n > t)\, dt$.

Step 2. Substitute $t = x^n$ ($dt = nx^{n-1}dx$); for $X, x \geq 0$, $X^n > x^n$ iff $X > x$.

Step 3. For $\mathrm{Exp}(\lambda)$,

$$E[X^2] = \int_0^\infty 2xe^{-\lambda x}\,dx = \frac{2}{\lambda^2},$$

so $\mathrm{Var}(X) = 1/\lambda^2$ (Section 5.5). ■

Exercise 5.8 (Variance of a bounded variable)

Let $P(0 \le X \le c) = 1$ for some $c > 0$. Show that $\mathrm{Var}(X) \le c^2/4$.

Solution.

Step 1. $X^2 \le cX$ on $\{0 \le X \le c\}$, so $E[X^2] \le cE[X]$.

Step 2. With $\alpha = E[X]/c \in [0,1]$,

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 \le cE[X] - (E[X])^2 = c^2\alpha(1-\alpha) \le \frac{c^2}{4},$$

as $\alpha(1-\alpha) = \frac{1}{4} - (\alpha - \frac{1}{2})^2$.

Step 3. Equality for $P(X = 0) = P(X = c) = \frac{1}{2}$. ■

Exercise 5.11 (Stein's identity and normal moments)

Let $Z \sim N(0,1)$ with density ϕ, and let g be differentiable with $E|g'(Z)| < \infty$, $E|Zg(Z)| < \infty$ and $g(x)\phi(x) \to 0$ as $|x| \to \infty$.
(a) Show that $E[g'(Z)] = E[Zg(Z)]$.
(b) Show that $E[Z^{n+1}] = nE[Z^{n-1}]$ for $n \ge 1$.
(c) Find $E[Z^4]$.

Solution.

(a)

Step 1. $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ satisfies $\phi'(x) = -x\phi(x)$.

Step 2. For $A, B > 0$, by parts,

$$\int_{-A}^{B} g'(x)\phi(x)\,dx = g(B)\phi(B) - g(-A)\phi(-A) + \int_{-A}^{B} xg(x)\phi(x)\,dx.$$

Step 3. As $A, B \to \infty$ the boundary terms vanish and the integrals converge ($g'\phi$, $xg\phi$ integrable); use Proposition 5.2.

(b) $g(x) = x^n$: all moments are finite ($|x|^k\phi(x) \le C_k e^{-x^2/4}$) and $x^n\phi(x) \to 0$, so $E[Z^{n+1}] = nE[Z^{n-1}]$.

(c) $E[Z^2] = 1$, $E[Z^4] = 3E[Z^2] = 3$; generally $E[Z^{2k}] = (2k-1)(2k-3)\cdots 1$, odd moments 0. ■

Exercise 5.29 (The probability integral transform)

Let X be a continuous random variable with distribution function F. Show that $Y = F(X)$ is uniformly distributed on $(0,1)$. (Example 5.9(d) assumed F strictly increasing; here F may be constant on intervals.)

Solution.

Step 1. F is continuous, nondecreasing, $F(-\infty) = 0$, $F(\infty) = 1$ (Proposition 4.1). Show $P(Y \le y) = y$ for $0 < y < 1$ and $P(Y \le 0) = 0$.

Step 2. For $y \in (0,1)$, $L = \{x : F(x) = y\}$ is nonempty (IVT), closed, bounded above ($F \to 1 > y$); let $q = \max L$, $F(q) = y$.

Step 3. $\{F(X) \le y\} = \{X \le q\}$: $X \le q$ gives $F(X) \le y$; $X > q$ gives $F(X) \ge y$ and $F(X) \ne y$ ($X \notin L$), so $F(X) > y$.

Step 4. Hence $P(Y \le y) = P(X \le q) = F(q) = y$.

Step 5. If $F > 0$ everywhere, $P(Y \le 0) = 0$. Otherwise let $a = \max\{x : F(x) = 0\}$ (nonempty, closed, bounded above); as in Step 3, $\{F(X) = 0\} = \{X \le a\}$, so $P(Y \le 0) = F(a) = 0$. ■